

BE International: Redeveloping an e- commerce application for the cloud

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ABOUT BE INTERNATIONAL

BE International redeveloped its B application on Gk support several k events, and posit company for bett revenue, and cus' experiences.



About BE International

Founded in August 2017, Beyond Eternity (BE) International is a direct marketing company whose e-commerce application, BE4U, enables sellers to market natural and holistic products in Malaysia, Singapore, Brunei, Hong Kong, and Indonesia.

Industries: Retail & Consumer Goods

Location: Malaysia

Google Cloud results

- Overcomes severe memory and compute limitations during high-transaction events
- Enables over 10,000 transactions per hour during flash sales or new product launches
- Reduces batch processing time to 30 seconds for 200 records
- Cuts down month-end commission calculations from 40 minutes to 2 minutes
- Frees up business

Imple
conta
to sup
sales



About Matrix Connexion

Matrix Connexion (MatrixC) has been a Google Cloud and Workspace Specialist Partner since 2008. It helps businesses unlock new capabilities, increase revenue, and save costs by adopting and migrating to the cloud.

Cloud Pub/Sub

Google Cloud (<https://cloud.google.com/>)

Specialized

BigQuery (<https://cloud.google.com/bigquery/>)

Google Workspace

Cloud Build (<https://cloud.google.com/build/>)

Cloud Functions

(<https://cloud.google.com/functions/>)

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

Monitoring (<https://cloud.google.com/monitoring/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine

(<https://cloud.google.com/compute/>)

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Container Registry

(<https://cloud.google.com/container-registry/>)

Looker Studio

(<https://cloud.google.com/looker-studio>)

technology
team to
focus on
development
roadmap
versus
infrastructure
maintenance

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Cloud IAM (<https://cloud.google.com/iam/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Operations
(<https://cloud.google.com/products/operations>)

Google Workspace
(<https://workspace.google.com/>)

Google Chat
(<https://workspace.google.com/products/chat/>)

Google Drive
(<https://workspace.google.com/products/drive/>)

Google Meet
(<https://workspace.google.com/products/meet/>)

When it comes to direct marketi
transact large sales volumes is
opportunity for companies. Bey
International (<https://beintl.com/>),
purveyor of naturally-derived prc
understands that facilitating a s
purchasing experience across ge
is key to rapid growth.

Founded in August 2017, the company's e-
commerce application, BE4U, was first built upon a
monolithic architecture. However, as the company's
membership base and monthly transaction volume
began to skyrocket, the application quickly met its
limits.



When success outpaces capabilities

There were three major events that would cause BE4U to shut down, which are flash sales, new product launches, and month-end transaction processing.

"It was chaotic when the system went down," explains Ben Chong, BE International's Head of Business Technology. "The project management staff had to manually generate invoices for successful transactions until past midnight whenever the system failed."

As a result, the BE International team received many complaints from its members, which jeopardized the company's image and sales revenue. It needed to break through these chokepoints, especially since it

planned to reach more members and open new markets in other territories.



Building a reliable application for large sales volumes

BE International's business technology team pinpointed four criteria to make BE4U a success:

- First, the application must be able to handle high traffic without compromising on performance especially, during its three major events.
- Second, it should enable its in-house team to solely focus on the development and enhancement of core functions instead of infrastructure hosting, maintenance, performance, and security issues.

- Third, BE4U needed to be more stable, robust, and resilient to ensure continuous operations.
- And finally, the team must be able to successfully reduce or minimize complaints from members while improving overall member and customer experiences.

Based on these criteria, Matrix Connexion (<https://www.matrixc.com/>) (MatrixC), a Google Cloud Partner, advised them to migrate BE4U to Google Cloud.

"We ultimately chose Google Cloud (<https://cloud.google.com/>) for its high service availability levels among competitors, regardless of the traffic volume or user location. Now that we are operating in Malaysia, Singapore, Brunei, Hong Kong, and Indonesia, the fact that fiber optic networks connect all Google Cloud data centers across the globe is a plus for us. Furthermore, the platform's pricing competitiveness can support our business growth in sustainable ways," reveals Chong.

He adds that reliable and fast connectivity enables the company's systems, websites, and apps to gain always-on availability for members, customers, internal users, and partners. Furthermore, support from MatrixC and the Google Cloud team have been instrumental in helping the business technology team understand, migrate to, and redevelop the BE4U application on its new microservices-based architecture.

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—Ben Chong, Head of
Business Technology, BE

The pivot from monolithic to microservices

The lift and shift from on-premise to Google Cloud was a major undertaking for BE International's IT lifecycle management and infrastructure modernization plans. The integration capabilities provided by Google Cloud has helped the 6-people integration team to deliver outsized gains.

"By using the CI/CD pipeline, Cloud Build (<https://cloud.google.com/build>) has helped us to accelerate from test to production within 5 to 10 minutes. This is a significant improvement in the application and path update process which now takes hours to days instead of weeks or months," Chong says.

The BE4U application was completely redesigned on the Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE) framework to host and manage its core systems. GKE has helped to ensure continuous high performance for the BE4U application in several major ways. For example, its auto-healing feature has helped the business technology team recover stopped services without manual interference around the clock. It also enables engineers to autoscale resources up and down dynamically in relation to user traffic or workload usage.

"This made our system even more resilient and highly available with optimum performance to serve our users better and more efficiently. It's easy to fix issues on the fly and roll out new features since the system is designed to perform with zero-downtime in mind," says Chong.

The ability to match workload with demand means the company is able to optimize the IT costs for its 50 microservices encompassing member, sales, payment, inventory, and bonus modules to support business growth, agility, and scalability. The company plans to add different services over time with GKE in line with future business needs, such as marketing or sales campaigns, and has the freedom to manage deployed services as and when needed.

It has also been able to realize tangible revenue gains from increased scalability, while the flexibility of a microservices architecture has opened new avenues for the company to build more features and upgrade existing features more regularly.

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Creating a smarter application for future growth

The partnership with MatrixC has also resulted in further enhancements to support on-cloud operations. In 2021, the company and MatrixC worked together to implement a new third-party backup integration with Google Cloud to protect the 80 percent of workloads currently running on the cloud. With its backup zone anchored in Taiwan, the company is able to backup workloads in all three countries it is operating in, giving BE International the peace of mind that data restoration will be easy.

Furthermore, while containerization has transformed BE4U, BE International's tech stack has been further enhanced with other Google Cloud solutions like Cloud SQL (<https://cloud.google.com/sql>), which serves as its transactional database, BigQuery (<https://cloud.google.com/bigquery>), for long-term analytics, and Cloud Operations (<https://cloud.google.com/products/operations>), for monitoring and alerts, among many others.

"With Google Cloud in place, we are able to achieve up to 60 times more performance for certain processes when compared to our initial monolithic approach," shares Chong.

Beyond technical gains, BE International is also benefiting from easy cross collaboration among departments with Google Workspace (<https://workspace.google.com/>), which enables round the clock and open communication with Meet (<https://workspace.google.com/products/meet/>). Files and access to documents are also securely stored on Drive (<https://workspace.google.com/products/drive/>). "Cross collaboration is important for us because BE International operates beyond just Malaysia, and we have partners and vendors across Asia," says Chong.

Prior to the transition to Google Workspace, BE International engaged MatrixC's professional services on the email migration and knowledge transfer to IT and all users. Upon completion of the transition, there is a continuous Google Workspace Enablement session for business users and key users. Here are some of sessions:

- **Business Transformation Workshop for business user and Head Of Dept**
- **Intermediate Google Sheets for Finance and Warehouse**
- **Google Forms & Google Sheets for Head of Dept and key users**

Outcome from the session, users from Finance started exploring Google Sheets, HR, Customer Service & Event started using Google Forms and reduction in dependence of Microsoft Office. Most of the users understand why transformation and innovation are important to their business and brainstorming areas of their business where Google can be applied was valuable.

Together, the results have transformed the BE4U experience. For instance, batch processing time is now down to 30 seconds for 200 records, compared to the previous time of 30 minutes. Sales bonus calculation time has been shortened to 2 minutes from the previous 40 minutes. These efficiency gains are part and parcel of longer-term strategies to make the company's deployment roadmap a success, which include plans to collate customer data, do deeper analysis, and retarget or personalize customer shopping experiences.

The future vision for BE4U is for it to support a highly configurable platform and evolve to be a superapp with a strong backbone, rich features, and diversified member experiences by 2024. As the company can now focus solely on stabilization and leveraging its data, it is exploring new ways for BE4U

to deliver naturalistic products and services through Google Cloud.

"The extent of Google Cloud solutions enables us to scale our applications and leverage machine learning or artificial intelligence in the future. We are worry-free about our deployment roadmap with Google Cloud as it is the most scalable and reliable infrastructure cloud platform," concludes Chong.

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